

Project Initialization and Planning Phase

Date	1 July 2026
Team ID	SWTID-2026-4523
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal Report:

The proposed solution aims to improve the credit card approval process using machine learning techniques. The system analyses applicant information and predicts whether a credit card application should be approved or denied. By automating the decision-making process, the solution reduces manual effort, improves prediction accuracy, speeds up approvals, and enhances customer satisfaction.

Project overview	
Objective	The primary objective of this project is to develop a machine learning-based system that predicts credit card approval using applicant information. The system aims to provide accurate, fast, and consistent approval decisions while reducing manual verification.
Scope	The project focuses on building a web-based application that analyses applicant details such as income, employment status, family status, education, and other relevant information. The trained machine learning model predicts whether the application is approved or denied, helping financial institutions improve efficiency and decision-making.
Problem Statement	
Description	Traditional credit card approval methods rely heavily on manual verification, making the process slow, time-consuming, and prone to human errors. As the number of applications increases, maintaining accuracy and consistency becomes more difficult.
Impact	An automated prediction system improves operational efficiency, reduces processing time, minimizes human errors, and provides faster and more reliable credit card approval decisions, leading to better customer satisfaction.
Proposed Solution	

Approach	The proposed system uses machine learning algorithms to analyse applicant data and predict credit card approval. Data preprocessing, feature engineering, model training, and prediction are performed to ensure reliable and accurate results.
Key Features	- Machine learning-based credit card approval prediction. - User-friendly web interface developed using Flask. - Fast and accurate approval or rejection prediction.

Resource Requirements:

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications, Number of cores	Standard CPU (Intel i5 or equivalent)
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	256 GB SSD or above
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Source, size, format	Kaggle Credit Card Approval Dataset